

A SHARED SCHOOL DAY.  
A CLEAR NEXT ACTION.

# School Operations Platform

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## *Product, Design and Engineering Blueprint*

Version 1.0 | 15 September 2026

Prepared for Abhishek and the people responsible for building, operating and validating the product.

Give every responsible adult the right next action, a shared understanding of the school day, and a reliable way to close the loop—without requiring the child or every parent to use a smartphone.

|                    |                        |                      |
|--------------------|------------------------|----------------------|
| 27                 | 40                     | 36                   |
| Blueprint sections | Prioritized work items | Acceptance scenarios |

### Decision status

A proposed product and SDLC baseline—not a production implementation, field-validation report, safety certification or legal opinion. Performance and commercial targets require measurement. The product name remains open; earlier project files use EDUVERA.

# Contents

Use the blueprint for alignment; use the companion specifications, contracts and scenarios for implementation.

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## How to use this pack

Read this document to align the team. Use the companion domain/contracts, workflow specifications, acceptance scenarios, backlog, architecture decisions, risk register and discovery/pilot runbook to turn alignment into implementation. The schemas and API contract are deliberately bounded examples for the first delivery slice, not an exhaustive implementation of every capability.

The earlier architecture already proposed a TypeScript modular monolith, native mobile plus operations web, PostgreSQL, scoped authorization, offline attendance and tool-mediated AI. Preserve those useful decisions unless an architecture decision record demonstrates a reason to change. [P1] This blueprint intentionally narrows delivery order around daily coordination rather than launching every module from the earlier broad product specification simultaneously. [P2]

## 1. Product definition and strategic position

Build an **India-first, configurable daily school-operations platform** connecting home, school staff and institution-managed services. The student is a person receiving education and care and the subject of operational records; a software account is optional. Adults and authorized devices perform digital actions. No core journey depends on a student phone.

The initial product connects four questions: What is expected today? What has actually been observed? Who needs to act? Has the outcome been confirmed? It coordinates preparation, attendance, day changes, communication, approved departure arrangements and selected handovers. It also supports staff coverage, basic care escalation and administrative actions affecting the day.

Position it as a school-operations system with closed-loop coordination, not as a surveillance system, an autonomous school administrator or merely a collection of registers. It may coexist with an existing school information or fee system. State which system owns which record.

A candidate differentiator is the combination of explicit expectations, contextual evidence, accountable handovers, non-app participation, safe configuration and low staff workload. That combination is a hypothesis to validate, not a claim of world-first or patentable novelty. Existing products already advertise attendance, fees, timetables, transport and parent communication; a longer feature list is not sufficient differentiation. [S1]

**Initial customer hypothesis:** Independent or small-group Indian day schools willing to designate an operations owner and pilot one daily workflow. Recruit schools with manual, mixed and existing-device practices, including households without smartphones. Do not assume a specific board determines all operational practices.

**Buyer:** School leadership or management. **Daily operators:** Teachers, office staff, transport attendants/coordinators and authorized care staff. **Beneficiaries:** Children and families. **Operational sponsor:** A named school leader with authority to change working practices.

## 2. Product constitution: non-negotiable principles

**Child-centered, not device-centered.** Use operational checkpoints only where they support care, coordination or a defined obligation. Do not record a child's entire day simply because it is technically possible. No advertising, sale of student data, covert monitoring, automated child-risk scores or public rankings of families.

**Observation is not unquestionable truth.** A bus location is not a child's location. A scanned credential is not verified collection authority. Missing data is not proof of absence. Every status must say what it establishes and retain its source.

**Authority is purpose-specific.** Receiving academic updates, paying fees, granting consent and collecting a child are distinct permissions. A phone number is a contact endpoint, not a complete identity or authorization model.

**Physical responsibility is not created by an app.** Model operational supervision and its recorded handovers separately from institutional accountability and legal guardianship. The software supports adults; it neither replaces supervision nor guarantees physical safety.

**Configuration must preserve invariants.** Schools can choose supported methods, owners and channels. They cannot configure cross-tenant access, fabricate observations, erase unresolved critical incidents or substitute a payment status for collection authority.

**Inclusive operation is part of the product.** Manual capture, assisted office processes and paper-backed evidence are valid first-class paths. No parent is required to purchase a smartphone to participate in the initial workflows.

**Reduce work rather than relocating it.** A record should be captured once and reused. Each new staff interaction needs an explicit operational benefit and an observed workload budget.

**Build a domain product, not a universal programming platform.** Keep school concepts and safety rules explicit in code. Configure bounded variation; add new domain capabilities through reviewed development.

### 3. Boundaries and staged capability map

| Capability                | First operational pilot                                                            | Next extension                                            | Deliberate boundary                                                             |
|---------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------|
| People and authority      | Student/staff import, enrollment, guardian links, pickup permissions               | Admissions conversion, group/campus changes               | No automatic merging by phone/name; no national identity requirement by default |
| Today and day plan        | Calendar, read/import timetable, materials, notices, operational changes           | Rich timetable editing and resource planning              | No continuous classroom monitoring                                              |
| Attendance and leave      | Teacher/manual/paper capture, late arrival, leave, corrections, discrepancy review | Approved gate/QR/Rfid/device adapters, session attendance | Gate and transport evidence remain separate                                     |
| Actions and communication | Action inbox, consent, announcements, threads, app/SMS/paper/call log              | Approved WhatsApp/voice integrations, advanced routing    | Delivery is not acknowledgment or consent                                       |
| Staff coverage            | Duty assignments and a simple substitution task                                    | Coverage planning and constrained scheduling              | Not payroll or employee-surveillance software                                   |
| Dismissal                 | Daily departure plan, approved changes, gate collection                            | End-to-end transport under a separate readiness gate      | No release based on cached authority alone                                      |
| Transport                 | Manual roster path only where part of the pilot                                    | Trip operations, vendor feeds, QR/Rfid, vehicle status    | No manufactured hardware; no independent-van guarantees                         |
| Care and safeguarding     | Restricted referral, owner, urgent contact and documented outcome                  | Specialist controlled workflows                           | Not diagnosis, clinical records replacement or autonomous safeguarding          |
| Fees                      | Link/read existing dues where necessary                                            | Fee subledger, online/cash/bank receipts, reconciliation  | Not a payment aggregator or full accounting ERP                                 |

| Capability     | First operational pilot                                           | Next extension                                        | Deliberate boundary                                          |
|----------------|-------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------|
| Academics      | Today requirements and simple assignment/notice links             | Assessments, results, richer assignments              | No full LMS/video/proctoring in the initial product          |
| Administration | Document request, lost item, basic office task                    | Admissions, certificates, library/hostel integrations | No unlimited generic form-builder project                    |
| AI             | Optional read-only, permission-scoped summaries after foundations | Reviewed drafting and bounded low-risk proposals      | Never authorizes release or independently resolves authority |

The full product can grow wider; the first usable release must be a complete narrow loop. Initially exclude payroll/HRMS, general accounting, a full video LMS, proctoring, biometric/GPS hardware manufacturing and a social network, consistent with the earlier specification. [P2]

Boarding, college self-service, independent transport management and complex campuses are extension domains, not settings that make them instantly supported. Model room for them, but advertise only tested arrangements.

## 4. Outcomes and commercial hypothesis

The primary outcome is **fewer unresolved daily coordination tasks with less staff effort and clearer family understanding**. Count confirmed resolution, not messages sent or screens opened.

Track median and tail resolution time, missing-register duration, staff minutes per class, parent clarification calls, percentage of required actions completed through non-app paths, authorization errors, duplicate records, and false or withdrawn alerts. Preserve baseline definitions and denominators.

Use a proposed north-star measure: **percentage of due operational obligations completed or explicitly resolved within their policy window**. An exception is not completed merely because someone dismissed its notification. Exclusions, excused obligations and manual overrides must remain visible so the denominator cannot be gamed.

Pair that measure with guardrails: no unresolved high-severity software-caused safety/security defects at release; no increase in unnecessary teacher workload; no deterioration in participation for non-app families; and no false claims of confirmed presence or handover.

A proposed pilot success hypothesis is at least a 20% reduction in staff coordination time for the chosen workflow, relative to a measured baseline, without weakening the guardrails. This is a test threshold, not an expected result or sales claim. Small pilots cannot establish rare-event safety statistically.

Commercially, test a school-paid campus/student subscription plus transparent integration and communication costs. Basic parent participation should not be paywalled. Safety-relevant records, approved offline arrangements and data export must not disappear abruptly because a messaging balance is empty or a subscription lapses. Define a contractual transition/read-only process rather than an unsafe sudden cutoff.

Measure onboarding labor, support effort, channel spending and integration maintenance per school. A profitable product needs repeatable configuration rather than custom software for every campus. Do not fix pricing or market-size claims before customer evidence.

## 5. People, roles and service ownership

| Role                               | Primary responsibility in the product                                               | Restricted by default                                            |
|------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Parent/guardian                    | Understand today, submit eligible requests, respond, receive child where authorized | Other children's records; permissions not explicitly granted     |
| Teacher/class teacher              | Attendance, day requirements, assigned students and classroom-related actions       | Unassigned classes and unrelated financial/care details          |
| Office/reception                   | Requests, verification, arrivals/departures, follow-up coordination                 | Unrestricted sensitive-case access                               |
| Transport attendant                | Assigned-trip rider observations and handovers                                      | Full academic/financial profile; unattended driving interactions |
| Transport coordinator              | Routes, staffing, expected riders, exceptions and trip closure                      | School-wide private case data                                    |
| Designated care/safeguarding staff | Restricted concerns, escalation and appropriate contact                             | Automatic publication of case content to routine feeds           |
| Principal/operations lead          | Exceptions, delegation, coverage, policy approval, school outcomes                  | Unlimited access merely because of job title                     |
| Accountant                         | Fee obligations, receipts and reconciliation                                        | Collection authority and confidential safeguarding records       |
| Implementer/support                | Configuration, training, diagnostics and time-bound approved assistance             | Standing access to all tenants or child records                  |
| Student                            | Subject, beneficiary and source of concerns through trusted routes                  | No required device or account in the core journey                |

Assignments need effective dates, scope, backups and revocation. One adult can have several roles; select the role/context deliberately and evaluate the requested action. Do not let being a teacher broaden that person's parent access elsewhere.

Distinguish institution-managed transport, contracted transport and family-arranged transport. Record what the school can observe and what the provider contract covers. Represent external observations as such rather than implying school-operated supervision.

## 6. Experience design and information architecture

Design around three questions: **What needs me now? What is happening today? What can I do?** Modules remain available, but daily users should not have to assemble the answer from separate attendance, messaging and transport screens.

For parents, use Today, Actions, Conversations and authorized Tools with a prominent child/school context. Multiple children can share a family overview without mixing permissions or tenant data. For teachers, prioritize My day, My classes, Attendance and Actions. For office staff, prioritize Arrivals, Requests, Dismissal and Unresolved cases. For attendants, expose only the assigned trip, riders, stops and exceptions. For leadership, show exceptions, workload and overdue decisions before decorative totals.

Keep navigation stable for a given permission set. Use a small number of persistent anchors plus the user's authorized high-frequency tools; place the remainder in Tools. Permission changes may change available tools, but ordinary events should not unpredictably reorder tabs. Desktop workspaces support denser queues and bulk operations; do not force desktop layouts into a phone.

Each operational card has a subject, plain-language status, time/freshness, responsible owner when relevant, and one clear next action. Separate information, required action, exception and confirmation. Avoid green checkmarks for planned or inferred events. Make urgent information understandable without color.

Design every screen's empty, loading, cached, pending sync, failed, disputed, corrected, unauthorized, disabled-module and archived states. An offline submission should say 'Saved on this device; school update pending,' not 'Confirmed by school.' A parent view can say 'Attendance not yet recorded' without suggesting the child is missing.

Use shared semantic design tokens for typography, surfaces, focus, severity, spacing and motion. Allow school branding only within accessibility constraints. Proposed interaction baseline: readable body text around 16 CSS pixels or native equivalent, large touch targets, clear contrast and restrained animation. Use WCAG 2.2 AA as the web acceptance baseline and apply equivalent accessible native patterns, tested with assistive technology. [S2]

The child does not need an app to be heard. Include an adult-assisted concern route and a restricted alternative route when a concern involves the ordinary handler. Do not automatically expose a confidential allegation to the subject of that allegation or every guardian.

## 7. Domain architecture: explicit modules with shared primitives

Use bounded domains, each owning its data, commands, permissions and invariants. Shared primitives reduce duplication; they do not erase the differences between money, attendance, consent and physical handover.

| Domain                    | Owns                                                                    | Examples of emitted business events                          |
|---------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------|
| Identity and access       | Accounts, memberships, role grants, authorization policies              | MembershipRevoked, AccessGrantChanged                        |
| People and enrollment     | Student identity, guardian relationships, enrollment, staff assignments | EnrollmentChanged, GuardianAuthorityChanged                  |
| School-day planning       | Calendar, timetable instances, duty coverage, daily expectations        | DayPlanPublished, DutyCoverageChanged                        |
| Attendance                | Sessions, observations, accepted attendance records, corrections        | AttendanceRecorded, AttendanceCorrected                      |
| Operational requests      | Leave, collection changes, consent decisions, dependency readiness      | RequestSubmitted, RequestApproved, ExecutionReadinessChanged |
| Dismissal and handover    | Departure plans, release authorization checks, recorded handovers       | DeparturePlanActivated, HandoverConfirmed                    |
| Transport                 | Routes, stops, trip legs, rider participation and closure               | RiderBoardingRecorded, TripClosureRecorded                   |
| Care and restricted cases | Referral, assigned owner, contact and case outcomes                     | RestrictedCaseOpened; minimal public envelope only           |
| Communication             | Threads, notices, delivery attempts, acknowledgment and response        | MessageRecorded, ActionAcknowledged                          |
| Finance extension         | Obligations, payments, allocation, receipt, reconciliation              | PaymentRecorded, ReceiptIssued                               |
| Configuration/integration | Policy versions, adapter mappings, provenance, connector health         | PolicyActivated, ObservationQuarantined                      |

| Domain           | Owns                                                                | Examples of emitted business events  |
|------------------|---------------------------------------------------------------------|--------------------------------------|
| Platform support | Documents, audit, reporting, job delivery and operational telemetry | ExportPrepared, SupportAccessGranted |

Use common concepts such as `PersonReference`, `ActorContext`, `EvidenceReference`, `Task`, `Obligation`, `PolicyVersion` and `CaseReference`. Keep a fee ledger distinct from an attendance register. Use relational models for important fields; limited JSON is suitable for validated configuration and extension metadata, not every domain record.

## 8. Core conceptual model and data invariants

The model distinguishes **Person**, **StudentEnrollment**, **Account**, **Relationship**, **AuthorityGrant** and **ContactEndpoint**. A student can exist without an account. A guardian can have one shared contact endpoint without having the same authority as every person using that phone. Keep tenant-local identities separate from optional global sign-in identity; do not merge people across schools by matching names or numbers.

An expectation specifies the subject, context, due window, expected outcome, responsible role, policy version and applicability. An observation states what a specific actor/device reported, when and in what context. A decision applies the appropriate policy to observations and authority. A domain event records an accepted business change. A projection is a derived view for a user; it is not the final authority for releasing a child or recording money.

A handover describes an expected/actual operational transition with authorized parties, required evidence and an outcome. A case captures unresolved or contradictory information with a responsible handler. A task is work to perform; completing a task does not automatically assert that a physical event occurred.

**Do not represent the entire child as one mutable status.** Maintain separate dimensions for participation expectation, classroom attendance, journey participation, departure arrangement, latest observed supervision and unresolved cases. A child may be present in class but not traveling on the afternoon bus. A latest supervision observation is not continuous real-time knowledge.

Store effective time and recording time where they differ. Important associations are date-bounded: enrollment, bus assignment, permission, role, policy and departure arrangement. Historical records retain the versions used. No silent overwrites of financial or safety-relevant corrections.

Primary invariants: records cannot reference another tenant's subject; only authorized commands create accepted decisions; one active departure-plan revision per student/day; one accepted current attendance decision per session/student; every correction links to what it supersedes; physical-release confirmation requires its own evidence; contradictions remain visible; neither channel delivery nor GPS arrival implies collection.

## 9. Event and observation model

Normalize inputs through a typed boundary: **capture -> authenticated observation -> validation -> reconciliation/decision -> business event -> task/communication/read model**. Adapters do not publish arbitrary 'student safe' facts.

An observation envelope includes tenant/campus scope, subject reference, semantic kind, context/session/trip, `observed_at`, `received_at`, `captured_by`, device/integration identity where applicable, source record ID, evidence references, offline status and schema version. Store consent/authority evidence in restricted records; do not copy sensitive documents into every event.

Accepted domain events additionally identify aggregate, aggregate version, correlation/causation IDs, applied policy version, actor and privacy classification. Use scoped deduplication on tenant, source and source record ID. The same ID with a different payload is a conflict, not a retry to overwrite. Use a schema version and documented compatibility policy.

A hardware credential scan can support an arrival observation under an approved mapping. It cannot automatically prove classroom attendance or collection authority. Do not rank methods through a universal confidence number: adequacy depends on context, provenance, spoofing risk, staff verification and freshness.

For asynchronous integration, a consistent event envelope is useful; CloudEvents is a reference for interoperable event metadata, not a substitute for our school-domain semantics. [S3] The provided initial envelope is a product contract, not a claim of complete CloudEvents conformance.

Keep audit/correction history protected for the approved retention period. 'Append-only' describes ordinary application behavior; it must not become an instruction to retain personal information forever or to defeat lawful deletion, redaction and legal-hold processes.

## 10. State machines and closed-loop workflows

Use explicit state machines for important workflows. Avoid one generic status enumeration used everywhere.

**Collection-change request:** Draft -> Submitted -> Verification pending -> Approved/rejected -> Operational readiness pending -> Ready -> Executed/cancelled/expired. Approval is a decision; readiness means responsible staff have the arrangement; execution means a separate physical event was recorded. Revocation can invalidate a future action even after earlier approval.

**Attendance:** Not recorded -> Observation pending -> Accepted present/absent/late or review required -> Corrected by an authorized process. An unsubmitted register is a staff task, not automatic student absence. Leave changes the expectation but does not manufacture a presence observation.

**Handover:** Expected -> Ready for verification -> Confirmed, or Exception open. Resolution of an exception records the actual outcome and may lead to confirmed handover, return to supervised school care, or another authorized outcome. A timeout never means success.

**Communication action:** Issued -> Delivery attempted -> Delivered where reported -> Acknowledged when observed -> Decision received where required -> Applied -> Closed. Many informational messages do not require all stages. Missing delivery information remains unknown.

**Critical reconciliation example:** A child has an accepted morning boarding observation but an unexplained conflicting classroom absence. Assign a responsible staff member and activate the school response. Do not wait for the ordinary absence batch or suppress the contradiction merely because the register arrived late. Record the subsequent investigation and corrected outcome.

The companion workflow specification defines normal, exception, offline and cancellation paths for six first-release workflows. Every additional workflow must satisfy the same contract before implementation.

## 11. Configuration system: flexibility with boundaries

Use three levels of change: **configuration** for supported parameter variation; **approved templates** for supported process variants; **development extensions** for new semantics or adapters. A new board, hostel process or transport contract is not automatically supported because a custom field exists.

Configuration may select actor roles, capture method, permitted evidence, operating windows, routine escalation, channels, language, grade/session policy and allowed request types. Platform invariants and

approved safety floors are not editable school settings. Material changes require authority, an effective time and a validation result.

Resolve policies by typed domain scope, not a universal 'most specific wins' tree. An attendance policy may resolve by school, campus and grade; a transport policy by service agreement, route and trip; a temporary exception by the specific student/date. Each policy type declares its eligible override dimensions and precedence. Equal-priority conflicts fail activation. Restrictive authority revocation cannot be overwritten by an ordinary convenience override.

Policy lifecycle: Draft -> Schema/type validation -> Dependency/invariant checks -> Scenario rehearsal -> Approval -> Scheduled activation -> Active -> Superseded/withdrawn. Keep version, author, approver, reason and affected populations. In-flight workflows reference a version, while current revocations and emergency restrictions are rechecked at action time.

Provide an explanation view: 'For this child today, manual class-teacher attendance applies because Grade II inherits Campus A policy v3.' Preview affected students, open workflows and message changes before activation.

Begin with fixed workflow templates and typed decision tables. No administrator-supplied JavaScript, SQL, unrestricted expressions, arbitrary loops or connectors with broad credentials. The initial policy schema in this pack is intentionally limited to attendance and associated communication/fallback settings; new policy types require review.

## 12. Manual, assisted, device and offline operation

Support a capability profile for each campus and workflow: available operators, devices, connectivity, installed integrations and non-digital paths. A school can use manual classroom attendance, QR at the gate and paper for some parent decisions simultaneously.

Manual and paper paths must preserve who observed, who entered, when the event occurred and what evidence was reviewed. Staff entry on behalf of a parent is attributed to the staff operator and the claimed authorizing person, not disguised as a parent-app action. A signed paper is evidence to assess, not automatic proof of identity.

Prefer offline-first observation capture for assigned staff: scoped cached roster, local device queue, stable operation IDs and visible pending state. Revalidate on synchronization. If an account was revoked or a roster changed, preserve the report in a review/quarantine path without silently granting it current authority.

Do not permit offline approval of new collection authority, guardian-right changes, refunds or permission administration in the initial release. Physical procedures must continue under an agreed school fallback. Distinguish a staff record of what physically happened from the system's permission to perform a future action.

Cached release permissions are inherently vulnerable to unseen changes. A signed cache proves where a record came from, not that it is still current. When required freshness cannot be established, staff use direct verification through the school-approved fallback, continue supervision and record the result. They must not obtain a green automated release from stale data alone.

Provide trip/class-scoped fallback printouts with generated time, revision, responsible owner and reconciliation space. Minimize personal data, control copies and define secure disposal. Office transcription after the fact must not generate misleading real-time notifications for events that occurred hours earlier.

For encrypted mobile structured storage, the Expo SQLite documentation supports SQLCipher but specifies that it is not available in Expo Go. Production offline security therefore needs an appropriate native build and testing, not a prototype-only assumption. [S4]

## 13. Solution architecture and technology baseline

Preserve the earlier recommended TypeScript stack as a starting decision, not evidence of the current repository's implementation. No repository audit was performed for this blueprint. [P1]

**Clients:** React Native/Expo for Android and iOS; Next.js for operations/parent web and a small public surface. Share contracts, validation and design tokens, not every screen. Parents can use responsive web or assisted/non-app paths; native installation is optional. A deployed app is not a prerequisite for a printed or office-entered decision.

**Application:** NestJS modular monolith with explicit domain interfaces. Nest modules support encapsulation and exports; use those boundaries deliberately and test forbidden dependencies. [S5] Keep the critical workflow logic deterministic and independent of notification or AI providers.

**Data:** PostgreSQL for authoritative transactional records, policies, tasks, inbox/outbox and audit metadata. Object storage for files. Redis may support queues, caching and rate limits, but is not the financial or handover source of truth.

**Asynchronous services:** Workers process outbox events, reminders, delivery attempts, imports and reports. Use separate priority lanes so a bulk notice or report cannot starve operational reconciliation. Realtime is a scoped update channel, not the state store.

**External boundaries:** Identity provider, messaging services, payment gateway, transport/hardware feeds and school-system imports. Each uses an adapter with a declared authority, schema, health state, retry policy and data minimization contract.

Start with one product backend and workers, not independent microservices for every module, Kafka, a graph database or a general workflow engine. Split a service only after measured load, fault-isolation or ownership requirements justify the cost. A dedicated transport ingest worker may be separated operationally without fragmenting the domain into dozens of services.

## 14. Transactions, concurrency and integration reliability

Commands pass authentication, tenant and record authorization, schema validation, current-policy checks and concurrency checks before changing business state. Critical authorization decisions read authoritative records, not eventually consistent dashboards or client caches.

Write the accepted state change and its outbox entry in one database transaction. Publish after commit. Workers are idempotent and maintain a consumer inbox or processed-event record. The transactional outbox pattern addresses the dual-write problem, but duplicate delivery remains a possibility and consumers must handle it. [S6]

For same-database safety dependencies, use an application service that coordinates domain interfaces in a transaction. For example, activating an early-departure plan also changes the local afternoon travel expectation. Do not rely on a later push notification to establish a critical invariant. Updates to an external transport provider remain an explicit readiness dependency with retry, human fallback and visible failure.

Use optimistic version checks for conflicting edits and database locking/constraints where a release or allocation needs serialization. A queue or Redis lock is not the sole enforcement mechanism. The same idempotency key with a changed request body returns a conflict. Two concurrent release requests must not both produce independently valid confirmations.

Accept that events can arrive late, twice or out of order. Store `observed_at` separately from `received_at`, preserve aggregate versions and reconcile causally related events. Offline observations can be recorded without silently reversing a later authorized decision.

Notification jobs reference current domain state and their triggering version. Revalidate stale reminders and cancelled requests before dispatch. Exactly-once delivery to a parent's phone is not promised; use stable message IDs, provider idempotency where supported and transparent attempt records.

Realtime clients fetch a scoped snapshot, subscribe to authorized changes and resynchronize from a cursor after reconnect. Provide targeted retry/backoff rather than polling every screen every thirty seconds. Critical action execution rechecks authority even after a fresh realtime update.

## 15. Tenancy, authorization and security

Tenant scope is established by authenticated membership, not trusted merely because a client supplied a tenant ID. Persist tenant ownership on business records and include it in important uniqueness and relationship constraints. Enforce tenant and relationship scope in APIs, jobs, exports, search, realtime channels and file access.

Use role permissions plus relationship/attribute checks. OWASP recommends least privilege, denying by default and validating permissions on every request; use these as testable requirements, not interface-only rules. [S7] A parent cannot access a different child by guessing an ID, changing a filter or opening an old attachment URL.

Use PostgreSQL row-level security as defense in depth. Test application database roles, job identities and pooled-connection context. PostgreSQL documents that superusers and BYPASSRLS roles bypass row security, while table owners normally do too; do not run the ordinary application under such a role. [S8]

Require stronger authentication for privileged staff, school-assisted recovery for disputed guardian links, scoped session/device management and prompt account revocation. Mobile biometric unlock of an adult's device is distinct from collecting student biometrics. Account recovery must not weaken the authority model.

Encrypt transport and appropriate stored data, use managed key/secret controls, scan attachments, validate types/sizes and issue short-lived scoped file access. Minimize sensitive content in logs and push previews. Do not assume that deleting a notification revokes a screenshot or an already-downloaded file.

Support access is time-bound, purpose-specific and auditable; production data is not routine developer test data. Prefer metadata diagnostics and explicit school authorization. Emergency access reveals only what the approved process requires and never acts as a universal 'force release' permission.

Use OWASP ASVS 5.0 as a verification catalog with versioned control references and risk-based applicability. Map mobile-specific controls separately; passing selected ASVS tests is not a blanket security certification. [S9]

## 16. Privacy, safeguarding and regulatory review

The notified DPDP Rules, 2025 have phased commencement. Their education and child-transport exceptions are conditional and purpose-limited, not a general authorization for unlimited monitoring. Map the school's and platform's roles, purposes and applicable provisions with Indian counsel before pilot; do not equate all provisions being notified with all being in force. [S10]

Maintain a dated legal/applicability register covering data processing, current and scheduled obligations, school/board/state requirements, transport contracts, safeguarding routes and communications-provider requirements. Recheck before launch and material policy changes. This blueprint does not settle disputed guardianship, legal custody or reporting duties.

Our design baseline is purpose limitation and minimization: no raw child biometric templates in the initial platform; no mandatory Aadhaar linkage; no face/emotion tracking; restricted care records; neutral

notification previews; and no automatic disclosure of sensitive allegations to all connected adults. A permitted integration still needs a reviewed purpose, vendor controls and non-biometric alternative.

Separate activity consent, personal-data processing permissions where applicable, collection authority and communication preferences. Withdrawal or revocation must propagate to future actions and channels without destroying historical records that must be retained.

Define category-specific retention, authorized correction/redaction, legal holds, deletion propagation, backup expiry and export. Preserve access-audit integrity without retaining complete sensitive payloads forever. Do not label a database's soft-delete flag as completed erasure.

Communication setup must include the provider's verified sender/template onboarding and channel-specific requirements. TRAI describes registered-header/content-template controls for commercial messaging; determine the applicable message classification and current provider rules rather than assuming every school message is exempt. [S11]

## 17. Financial and administrative extensions

The initial product may integrate or link existing fee information. When implementing fee collection, keep a proper fee subledger: obligation, line items, concession authority, payment, allocation, receipt, refund and reconciliation are different objects. Use integer minor units or fixed decimal amounts with currency; never floating-point balances.

A client success screen is not sufficient payment evidence. Verify provider callbacks, reconcile with provider/bank records and account for duplicate and out-of-order delivery. Razorpay's documentation explicitly describes duplicate webhooks and event IDs for deduplication. [S12] Provider capture, bank settlement and allocation to a student's dues remain distinct states.

Cash and bank transfers need authorized entry, evidence, reconciliation and receipt controls. Changes use traceable adjustment/reversal processes, not silent deletion. Permission to refund is not the same as permission to collect or view fees.

Unpaid fees must not become a hidden condition in safety handovers, emergency communications or essential access to a family's own operational records. Financial policy is a separate reviewed business and legal matter.

Admissions can initially use verified import/manual enrollment. Later add enquiry, application, document review, offer and idempotent enrollment conversion. Add age/board-specific academic behavior deliberately; a preschool workflow is not a senior-school assignment workflow with different labels.

Keep resource management bounded: staff coverage, classrooms, vehicle capacity, activity capacity and approvals affecting daily plans are in scope as needed. Full inventory, facilities maintenance, meal systems and library/hostel systems can connect through tasks and interfaces before becoming native modules.

## 18. AI and product intelligence boundaries

AI is optional and must not sit in the critical path of attendance capture, permission checks, release, communication delivery or emergency response. The platform must remain fully operable when the model provider is unavailable or disabled.

The first useful AI capabilities are permission-scoped retrieval, evidence-linked day summaries, drafting routine notices and explaining approved configuration. It may extract a proposed request from a message, but the user must verify the structured request before submission. Treat incoming documents and messages as untrusted content, not instructions for tools.

All AI access goes through the same authorized application interfaces, with restricted data selection and logged tool use. Never give a model unrestricted database credentials. Read results identify sources and freshness. A text explanation is not an authorization record.

Do not use AI to identify guardians, authorize collection, diagnose a child, determine safeguarding outcomes, silently set biometric thresholds, infer sensitive traits or mark unexplained attendance discrepancies resolved. Human confirmation alone is not sufficient for every risky action; some actions are outside the AI tool set entirely.

Later bounded write proposals need explicit preview, eligible permissions, fresh server validation and deterministic execution. Red-team cross-tenant retrieval, prompt injection, misleading tool output, fabricated citations and stale plans. Use synthetic test cases and measure useful time saved rather than adding a chatbot as the primary interface.

A more distinctive near-term capability is **policy rehearsal**: run synthetic and approved de-identified historical scenarios through a candidate configuration and explain changed outcomes before activation. Begin with deterministic rules. This is a product hypothesis to test, not a claim of novel academic research.

## 19. Quality attributes and measurable engineering targets

These are **proposed targets for the reference workload**, not measured performance, customer SLAs or physical-safety guarantees. Final thresholds require testing on an agreed deployment and representative Indian networks/devices.

| Attribute              | Proposed acceptance target                                                      | Measurement boundary                                                             |
|------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| API responsiveness     | p95 below 700 ms for ordinary scoped reads/writes                               | Backend time, excluding large imports/reports and external providers             |
| Roster interaction     | A cached roster remains usable offline; online submission p95 below 2 s         | Defined 40-student roster on agreed device/network profile                       |
| Reference burst        | 2,500 teacher rosters of 40 students over 5 minutes, plus 500 reads/s           | 100,000-student synthetic fixture; test peak and fivefold short burst separately |
| Operational update lag | p95 below 5 s from accepted event to online operational read-model/task update  | Excludes disconnected devices and external carrier delivery                      |
| Availability           | Initial planning objective of 99.9% monthly for critical backend routes         | Define eligible requests and exclusions before a contract                        |
| Recovery               | Target RPO up to 5 min and RTO up to 60 min for regional disaster               | Must be proven by restore drill; process restart must not lose committed records |
| Tenant isolation       | Zero unauthorized cross-tenant disclosures in release test suite                | Include API, cache, files, search, jobs, realtime and AI                         |
| Duplicate safety       | Replayed commands/events do not duplicate a release, receipt or accepted record | Test changed-payload conflicts and concurrent attempts                           |
| Accessibility          | WCAG 2.2 AA for web and equivalent native task testing                          | Include keyboard/screen reader, errors and non-color status                      |
| Offline clarity        | All pending, stale and conflict states visible                                  | No unacknowledged local write shown as server-confirmed                          |

An RPO is not permission to silently lose a child's handover history. Recovery procedures must identify the affected time range, reconcile with physical/printed records and make uncertainty visible. Do not resume release decisions from an unverified stale recovery snapshot.

Instrument technical latency alongside domain health: oldest unprocessed observation, missing duty owner, unresolved rider count, stale plan revisions, disconnected devices, overdue calls and failed communication routes. No telemetry event should require unnecessary child identity.

## 20. Discovery and validation before code expansion

Run field discovery with approximately six to eight schools spanning manual, hybrid, existing devices, different shifts and transport ownership. This is a recruitment plan, not a representative national sample. Include non-app guardians, teachers, attendants, office staff and someone responsible for safeguarding.

Observe an arrival and dismissal period and a classroom attendance task. Map what staff actually do, not only the documented policy. Obtain permission and use de-identified notes; do not photograph child records casually. Record workarounds, repeated data entry, decision authority, failure cases, language and connectivity constraints.

Create a service blueprint separating child activity, parent touchpoints, visible staff tasks, backstage office work, systems and physical evidence. Run event-storming workshops for business events, commands, actors and ambiguities. Use an exception workshop to challenge missing receivers, wrong buses, late uploads and disputed links.

Compare competitors through the same scenarios, not a checklist of module names. Ask for demonstrations of offline/no-parent-app collection changes, authority revocation, contradictory attendance and traceable closure. Vendor marketing is evidence of a claimed feature, not verified workflow quality. [S1]

Test paper prototypes and realistic mobile flows with the actual operators before building complex automation. Measure completion time, error, comprehension, recovery from a mistake and assistance needed. Revise the scope if the workflow adds substantial work or depends on nonexistent roles.

Produce an evidence log where each hypothesis has a source, strength, open question and decision. The team should be able to distinguish an observed need, an assumed need and a proposed solution.

## 21. SDLC, traceability and release gates

Use iterative discovery and vertical delivery, not a long document followed by a single big-bang implementation. For each slice, connect **observed problem -> outcome -> requirement -> domain rule -> design -> contract -> code -> test -> release evidence -> pilot metric**.

Give artifacts stable IDs: hypotheses HYP, requirements REQ, workflows WF, invariants INV, decisions ADR, risks RISK and scenarios AT. Keep the mapping in version control. Critical policy changes deserve the same review discipline as code changes.

| Gate                          | Required evidence                                               | Accountable decision                                   |
|-------------------------------|-----------------------------------------------------------------|--------------------------------------------------------|
| G0: Problem and scope         | Observed workflow, baseline, operator and non-app path          | Product lead with school sponsor                       |
| G1: Domain and hazards        | Vocabulary, authority model, invariants, exception routes       | Technical lead and independent safety/privacy reviewer |
| G2: Interaction and contracts | Tested prototype, acceptance examples, schema/API design        | Design lead and engineering lead                       |
| G3: Implementation ready      | Reviewed code, migration, tests, observability and rollback     | Engineering lead and QA                                |
| G4: Shadow pilot              | Parallel comparison, staff drill, import/authority verification | School operations owner and release lead               |

| Gate                    | Required evidence                                                    | Accountable decision                               |
|-------------------------|----------------------------------------------------------------------|----------------------------------------------------|
| G5: Assisted live slice | Monitoring, direct support, reconciled exceptions, stop criteria     | School sponsor and release authority               |
| G6: Repeatable rollout  | Proven onboarding, cost/support metrics, restore drill, known limits | Product owner with engineering/operations sign-off |

NIST's SSDF provides a reference for integrating secure practices into development and supply-chain work. Use it to organize the team's secure-development evidence rather than postponing security until release. [S13]

Definition of Ready: actor, scope, authority, normal/exception/offline paths, data classification, acceptance tests and measurable benefit are known. Definition of Done: code and contract checks pass; accessibility and authorization are tested; runbook and migration exist; user copy handles uncertainty; telemetry is privacy-reviewed; no unresolved blocking risk remains; reviewer and product acceptance are recorded.

## 22. People, decision rights and collaboration

| Workstream                 | Accountable owner                    | Required collaborators                             | Main artifacts                                        |
|----------------------------|--------------------------------------|----------------------------------------------------|-------------------------------------------------------|
| Product scope and outcomes | Product lead/founder                 | School sponsor, researchers, customer success      | Charter, evidence log, priorities, outcome measures   |
| School-process correctness | School operations owner              | Teachers, office, transport and care leads         | Service blueprint, SOPs, escalation and fallback      |
| Experience and inclusion   | Product designer                     | Operators, non-app guardians, accessibility tester | Role flows, screen states, tokens, usability evidence |
| Architecture and data      | Technical lead                       | Backend/mobile/web engineers, privacy reviewer     | Domain model, ADRs, contracts, migrations             |
| Backend and integration    | Backend lead                         | Vendor contacts, QA, operations                    | Commands, policies, adapters, reconciliation          |
| Mobile and web             | Client lead                          | Design, backend and device testers                 | Role interfaces, offline queue, realtime recovery     |
| Verification               | QA/release lead                      | Engineers, school testers, security reviewer       | Tests, hazard scenarios, regression evidence          |
| Security/privacy           | Named security/privacy owner         | Indian counsel, school safeguarding lead           | Threat model, access matrix, applicability register   |
| Deployment/reliability     | Platform/operations owner            | Engineering, support, school coordinator           | CI/CD, monitoring, restore/rollback drills            |
| Adoption and support       | Implementation/customer-success lead | School champions, product and operations           | Training, import sign-off, support and offboarding    |

A small team may combine roles, but every decision still needs an owner. Separate the implementer from the reviewer for authority, financial and high-risk release changes. School acceptance does not replace software security review; software approval does not replace the school's operational readiness.

Work in short increments with regular school walkthroughs. Publish a release note stating behavior changes, affected roles, known limitations and fallback. Never silently change collection or attendance semantics through a routine UI deployment.

## 23. Verification strategy and difficult scenarios

Use unit tests for state transitions and authorization predicates; property-based tests for invariants; database integration tests for tenant constraints and races; API/adaptor contract tests; end-to-end role tests; accessibility tests; device/offline tests; and operator acceptance in a school-like environment.

Prioritize combinations: a non-app guardian, a shared phone, a revoked collector and an offline attendant; a teacher who is also a parent; a student missing morning transport but using afternoon transport; a late import after a corrected register; a cancelled request whose SMS is still queued; and a school network outage during dismissal.

Test adversarial input: copied QR credentials, duplicate source IDs with changed payloads, forged provider callbacks, client-supplied tenant IDs, direct calls to hidden APIs, signed-URL reuse, unauthorized export and prompt injection into AI retrieval.

Test semantic failures as well as technical failures: bus arrival mistakenly shown as student arrival, a staff-entered paper decision attributed to a parent account, an expired approval shown as ready, a notification viewed as consent, and a process timeout shown as completion.

Every critical scenario needs an expected human response and a software outcome. For example, a stale offline authorization must not yield automatic confirmation; the UI routes to direct verification, preserves supervision guidance and records the eventual factual outcome separately.

Use synthetic fixtures and controlled drills rather than unsafe real-world experiments. Stop a pilot for unauthorized disclosures, wrong-person association, false release confirmation, unowned critical cases or loss of essential fallback. A low count of real incidents is not evidence that all rare failure paths are safe.

## 24. Deployment, onboarding and operations

Create isolated development, staging and production environments. Use synthetic data in development. Automate linting, type checks, contract validation, tests, dependency/secret scans, reproducible builds and migrations. Store secrets outside code and narrowly scope deployment identities.

Release through feature flags by campus/workflow. Use expand-and-contract database migrations, migration rehearsal, backups and rollback/forward-fix plans. A policy rollback applies prospectively and must not erase events that physically occurred under the previous version.

Onboard through a repeatable checklist: service-boundary agreement, data/authority cleanup, source-of-truth map, import rehearsal, staff permissions, fallback contacts, channel setup, device provisioning, training, shadow period and sign-off. Clean roster and guardian data before relying on automation.

For the first live rollout, limit to one grade/section or one controlled transport route. Run alongside an agreed existing process in shadow mode before staff rely on digital decisions. Expand only after discrepancy review, staff competency and working fallback are demonstrated.

Operations need dashboards for queue lag, provider failures, disconnected devices, unresolved cases, stale plans and access anomalies. Separate technical on-call from the school's physical incident response. A child-care concern goes to the designated school response, not a generic software support ticket.

Offboarding includes usable exports with definitions, open-case handover, revoked accounts/integrations, device/cache handling, retained-record obligations and documented deletion schedule. A school should not need to remain a customer merely to obtain its own permitted data.

## 25. Roadmap: deliver in dependency order

**Stage A - Discover and specify the daily loop.** Validate the target school arrangement, student/guardian identity, vocabulary, authority, staff workload and physical fallback. Exit with G0-G2 evidence for the first slice, not just attractive screens.

**Stage B - Establish a usable coordination core.** Implement tenancy, memberships, people/enrollment, day plan, manual/offline attendance, leave, action inbox, notices/conversations, non-app office capture, scoped documents and audit. Deliver a complete attendance-to-parent-follow-up loop in one school context.

**Stage C - Controlled departure coordination.** Add daily departure plans, approved collection changes, staff readiness, restricted care/early-departure integration and gate handover. Pilot transport only with a separately verified roster, operator and exception procedure. Do not sell a full safety loop before this gate passes.

**Stage D - Repeatability and approved adapters.** Add selected QR/Rfid/biometric event integration, contracted-transport reconciliation, fuller staff coverage, provider observability, configuration rehearsal and onboarding templates. Validate each integration rather than claiming universal device compatibility.

**Stage E - Financial and academic breadth.** Expand fee subledger/reconciliation, admissions, timetable authoring, assignments/results and selected administrative workflows based on demand and measured operational value.

**Stage F - Intelligence and new segments.** Add bounded AI assistance and optimization after trustworthy records and permissions exist. Evaluate boarding, college and larger enterprise arrangements as separate product increments.

Do not commit dates from this document alone. Estimate after spike results, chosen staffing, integrations, legal review and pilot scope. The attached backlog contains dependencies and release intent, not invented effort estimates.

## 26. Immediate decisions and unresolved questions

Approve the product constitution, first customer hypothesis and narrow daily-loop release intent. Assign the accountable school-process, technical and privacy/safety reviewers. Choose discovery partners and select a low-tech and mixed-method scenario for the first prototype.

Resolve before implementation: school-managed versus contractor transport boundary; who verifies guardian changes; how long each offline cache may be used; who owns physical exceptions; authoritative student/enrollment source; exact channel/provider costs; and what staff can realistically record during peak activity.

Resolve before live collection workflows: fresh-authority method during outages, incident routing, authorized independent departure policy for eligible students, handling of disputed family authority, route/provider readiness and school-specific staff training.

Resolve before scale: actual peak workload, tenant isolation tests, recovery results, integration support effort, category-specific retention, legal applicability and achievable paid onboarding/support economics.

The best first demonstration is one complicated but realistic day executed correctly: absent teacher, child misses the morning bus, paper-backed collection change, delayed sync and a revoked receiver. A beautiful happy-path screen is useful; a correct and understandable recovery path is the stronger product evidence.

## 27. Final product definition

**A child-centered, adult-operated school coordination platform that turns daily plans into accountable actions and verified outcomes, across digital and non-digital participation, while making uncertainty and operational boundaries explicit.**

The guiding implementation rule is: **keep domain truth and authority precise; keep school variation bounded; keep the human experience simple.**

## References and provenance

External references were checked on 15 September 2026. Product recommendations are not presented as claims from these sources. Recheck provider specifications, versions and legal applicability before implementation or procurement.

[P1] Earlier project document, 02\_TECHNICAL\_ARCHITECTURE.md, retrieved from the user's Library. Recommended architecture, not a repository audit.

[P2] Earlier project document, 03\_PRODUCT\_FUNCTIONALITY.md, retrieved from the user's Library. Broad capability map and initial exclusions.

[S1] Fedena, official feature tour. <https://fedena.com/feature-tour>

[S2] W3C, Web Content Accessibility Guidelines 2.2. <https://www.w3.org/TR/WCAG22/>

[S3] CloudEvents project. <https://cloudevents.io/>

[S4] Expo, SQLite and SQLCipher documentation. <https://docs.expo.dev/versions/latest/sdk/sqlite/>

[S5] NestJS, Modules. <https://docs.nestjs.com/modules>

[S6] AWS Prescriptive Guidance, Transactional outbox pattern.

<https://docs.aws.amazon.com/prescriptive-guidance/latest/cloud-design-patterns/transactional-outbox.html>

[S7] OWASP, Authorization Cheat Sheet. [https://cheatsheetseries.owasp.org/cheatsheets/Authorization\\_Cheat\\_Sheet.html](https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html)

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[S9] OWASP, Application Security Verification Standard. <https://owasp.org/www-project-application-security-verification-standard/>

[S10] MeitY, Digital Personal Data Protection Rules, 2025, Gazette, especially rule 1 and Fourth Schedule. <https://www.meity.gov.in/static/uploads/2025/11/53450e6e5dc0bfa85ebd78686cadad39.pdf>

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[S16] JSON Schema Draft 2020-12. <https://json-schema.org/draft/2020-12>